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LESSON 4

Cost-Based Modeling

Learning Objectives

After completing this lesson, the student should be able to:

1. Specify and explain an appropriate model for mass appraisal using the cost approach
2. Use sale prices to develop a depreciation factor for the building portion of the cost model
3. Transform variables to make use of the cost data provided in a building replacement cost calculation
4. Calculate depreciated replacement cost new (RCN) for a property database
5. Create appropriate market adjustment factors for variations in building characteristics
6. Calculate a cost approach value by adding market adjusted building value to previously developed land value
7. Determine and apply needed adjustments for general factors such as location

ASSIGNED READING

1. UBC Real Estate Division. 2020. *BUSI 444 Course Workbook*. Vancouver, BC: UBC Real Estate Division.
Lesson 4: Cost-Based Modeling
2. UBC Real Estate Division. 2019. *Advanced Computer Assisted Mass Appraisal*. Vancouver, BC: UBC Real Estate Division.
Chapter 5: The Cost Approach
Chapter 6: Specifying and Calibrating the Cost Model



Selected readings can be found under Online Readings on your Course Resources webpage. Please note that recommended readings are not tested on the exam, but are provided for further research into these topics.

INSTRUCTOR'S COMMENTS

The cost approach to value has been criticized in appraisal literature as being of limited use. However, this criticism is primarily from a single property appraisal perspective. Single property appraisers generally do not maintain the amount of property information that is normal for an assessment jurisdiction. With this additional data, assessors can develop market-derived depreciation, which makes the cost approach much more effective. The cost method may also allow assessors to include more factors that buyers consider valuable in the valuation process. Because some factors, such as shake roof, porches, and patios have a low percentage of value, these may not be included in a multiple regression model, but can be included in a cost model. This may allow cost models to be more effective for assessment work than some market models.

Many researchers report the cost model structure is one that fits universally in many jurisdictions. The cost model structure is mentioned in the later lesson on non-linear regression as the basis for that hybrid model. In cases where there are not enough valid sales to use a regression to develop the coefficients in the model, cost manuals can be used with the final result tested and adjusted using what sales evidence is available.

Database for this lesson: we will be continuing with the “444costdata” database from Lesson 3. If you wish to work with a fresh database, you can download the “444costdataLesson4” database from the course webpage. However, this is not required, since this is the same database we saved at the end of Lesson 3 – you can simply continue with the version you saved.

STEPS IN THE DATA ANALYSIS PROCESS

The steps in producing the model are the same as those outlined in Lesson 3. These steps are reproduced below for your reference:

1. Describe an appropriate general model to use and state this model using standard mathematical symbols.
2. Review the variables in the database and identify those which are suitable for this type of model.
3. Examine the potential independent variables for relationships with each other and with the dependent variable using graphic analysis, cross tabs, and correlation analysis. (Please note: this step – as well as Steps 4(b) and 5 below – are required only for regression-based models and not for cost models.)
4. (a) Create any transformations necessary to make variables suitable for the chosen model structure.
(b) Create any additional transformations required to remove problems of collinearity identified in Step 3.
5. Repeat Step 3 with new variables.
6. List a final group of potential variables for calibration.
7. Calibrate the model using an appropriate method
8. Test and evaluate the model for use.
9. State your conclusions as to model quality.

Step 1 – Describe an Appropriate General Model to Use

The cost approach uses a hybrid model structure that contains both additive and multiplicative terms. Because of this model structure, analysis of possible relationships among the variables is not needed as it would be in additive regression analysis. The general cost model is as follows:

$$MV = \Pi GQ \times [(\Pi BQ \times \Sigma BA) + (\Pi LQ \times \Sigma LA) + (\Pi OQ \times \Sigma OA)]$$

where:

MV = market value

ΠGQ = the product of the general qualitative components

ΠBQ = the product of the building qualitative components

ΠLQ = the product of the land qualitative components

ΠOQ = the product of the other building qualitative components

ΣBA = the sum of the building additive components

ΣLA = the sum of the land additive components

ΣOA = the sum of the other building additive components

The land portion ($\Pi LQ \times \Sigma LA$) of the model was developed in the land valuation lesson. This lesson will concentrate on the development of the building component and the final market adjusted cost value.

Step 2 – Identify Variables Suitable for the Model

The variable selection process for the cost approach is somewhat different than the process under other model specification methods. To a large extent, the cost manual to be used determines the variables required because only variables that have cost data available in the cost manual can be used to determine the value of the building's additive features.

General Qualitative Variables

General qualitative variables are multiplicative adjustments for factors that influence value but are not accounted for in the building or land portion of the cost model. The only general qualitative variable used in this cost model will be neighbourhood. The time adjusted sale price developed in Lesson 3 will be the dependent variable against which we will test our predicted cost value. During the calibration step we will determine the need for a neighbourhood adjustment.

Building Qualitative Variables

Building qualitative variables include items such as condition, maintenance, architectural style, and depreciation. In this database, the main variables that reflect architectural style and quality for the main building are:

- $linmcls$ – linearized manual class. This variable is derived from the cost manual and relates the various quality and architectural styles to the base style of the 140 class;
- $mbmclsdv$ – main building manual class deviations. This variable adjusts the main class of the property for minor percent variations in quality and style from the predominant class it usually ranges from -15 to +15;
- $linmcdv$ – linearized manual class with deviations. This variable combines the previous two variables;
- $splitlvl$ – split level building style;
- $baseste$ – the home has a basement suite; and
- heating type – the predominant type of heating installed in the house.

In this lesson, we will use the following building qualitative variables:

- $linmcdv$;
- heating type (an adjustment for superior quality heating); and
- a newly created variable named $PCTGOOD$ which will represent all forms of depreciation.

Building Additive Variables

These variables are determined by what is available in the cost manual. The main building additive variables that are suitable to be included in this model are:

- $flrarea1$ – first floor area;
- $flrarea2$ – second floor area;

- bsmtarea – basement (full height) area;
- crwlarea – crawl space area;
- bsmtfin – finished basement area;
- baths – count of bathrooms; and
- fireplcs – count of fireplaces.

Other Building Variables

There are also variables that reflect the quality and style of the other buildings. These are:

- cp_manc1 – carport quality class;
- gr_manc1 – garage quality class; and
- gar_type – garage style

The other building additive variables that may be suitable for use in the model are:

- cp_area – carport area; and
- gar_area – garage area

The outbuildings do not have quality or area variables available, so ob1mktvl, outbuilding market value estimate, will be used to represent the value of these structures.

In Lesson 3 we developed the variable neighland; it will be used to represent the land portion of value.

The cost model does not require any analysis of the relationships among the variables, as model specification does not rely on any specific relationships between variables, but simply uses the values from the cost manual. Steps 3 and 5 will therefore be omitted.

Step 4 – Transform Variables to Suit Model Structure

Depreciation

Many cost manuals include age/life depreciation tables to determine depreciation factors under the cost approach. These have not been included in the data given in this lesson. The Replacement Cost New (RCN) and the Replacement Cost New Less Depreciation (RCNLD) variables in the database could be used to determine the depreciation assigned by the original appraiser who created these values. A high degree of consistency in the depreciation assigned may be an indication that age/life method may have been used. However, using the RCN and RCNLD variables would not allow you to state with any certainty which method of depreciation had been used previously. As it is normally considered more appropriate in a market adjusted cost approach to use market-derived depreciation rates, a method for developing these rates will be demonstrated in this lesson.

The first step in this process is to develop a market derived depreciation factor. This process will start with the creation of an improvement residual, which will be the time adjusted sale price less the neighland value developed in Lesson 3. The observed depreciation factor is found by dividing the improvement residual by the sum of all the replacement cost new (RCN) components (i.e., main building, garage, carport, and outbuildings). This will create a ratio representing all forms of depreciation. In order to analyze the depreciation, we need an age variable; a new variable effage will be used for this.

The following transformations were used:

COMPUTE imprrsid = tasprice – neighland.

COMPUTE obsdeprn = 1 – (imprrsid/(mb_rcn+cp_rcn+gr_rcn+ob1mktvl)).

COMPUTE effage = 2017 – mbeffyr.

The first transformation creates the improvement residual variable, the next creates the observed depreciation. As the residual in most cases will be less than that of the sum of the RCN values, the term in the brackets will be less than 1.00. Subtracting this ratio from one will create an amount of depreciation shown as a decimal (as opposed to a percent). The last transformation creates an age variable using effective year built.

We will use the Curve Estimation procedure in SPSS to find the best fit for the depreciation. Curve estimation will provide us with different curves to choose from. In our case, because market depreciation is rarely a straight line relationship, we show only the quadratic and cubic curves. The best representative curve should have the highest R^2 combined with a close to zero depreciation for new buildings (age zero). Also, the curve should generally not exceed 45% to 50% for total depreciation of the oldest buildings, since homes beyond this percent depreciation start to become uninhabitable. Most owners and landlords will undertake periodic maintenance and repairs that tends to stabilize the effective depreciation at less than 60% for older homes.

Use Analyze → Regression → Curve Estimation, place obsdeprn as the Dependent variable and effage as the Independent variable, choose Quadratic and Cubic under Models. Click OK.

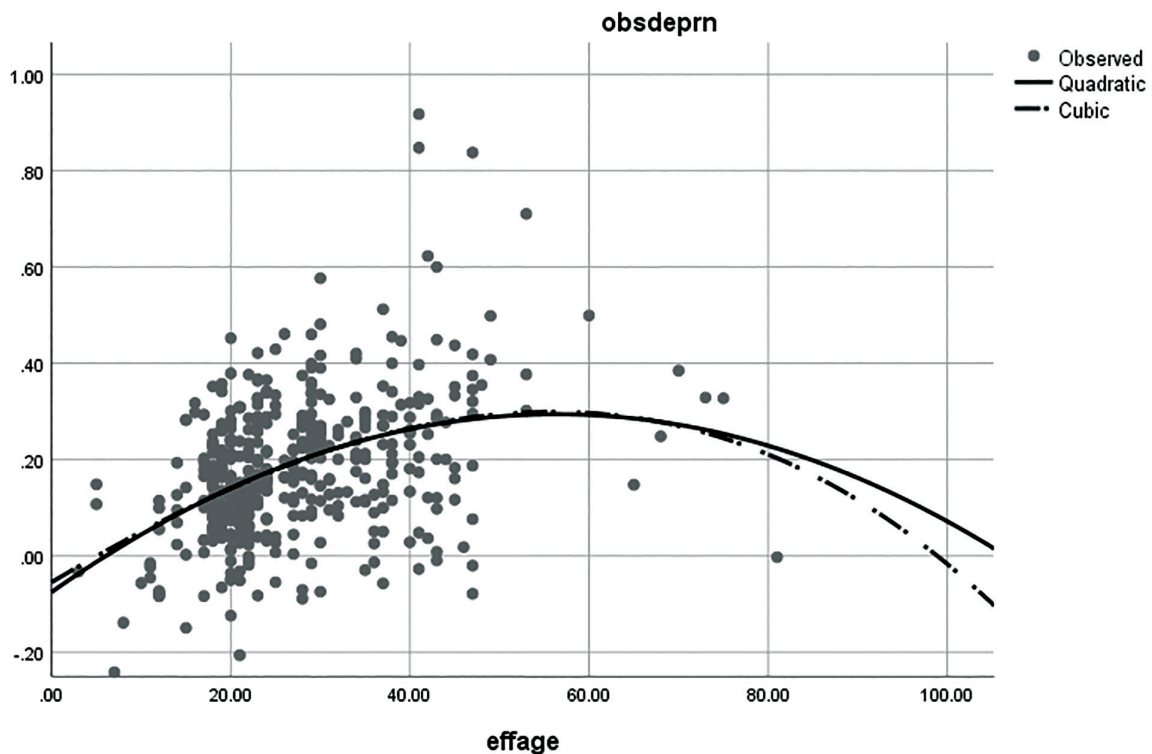
Your output should include the following table and chart.

Model Summary and Parameter Estimates

Dependent Variable: obsdeprn

Equation	R Square	Model Summary				Sig.	Parameter Estimates			
		F	df1	df2			Constant	b1	b2	b3
Quadratic	.169	45.471	2	447		.000	-.075	.013	.000	
Cubic	.169	30.296	3	446		.000	-.054	.011	-5.650E-5	-5.021E-7

The independent variable is effage.



The quadratic and cubic curves are both very smooth, but drop off too dramatically at the high end, after age 60, implying homes older than 60 years will have less depreciation than newer ones. This is not logical. To account for this, we will create a new variable effage2 that will cap our effective age at 60. That is, convert all ages over 60 to age 60.

COMPUTE effage2=effage.

IF(effage>60)effage2=60.

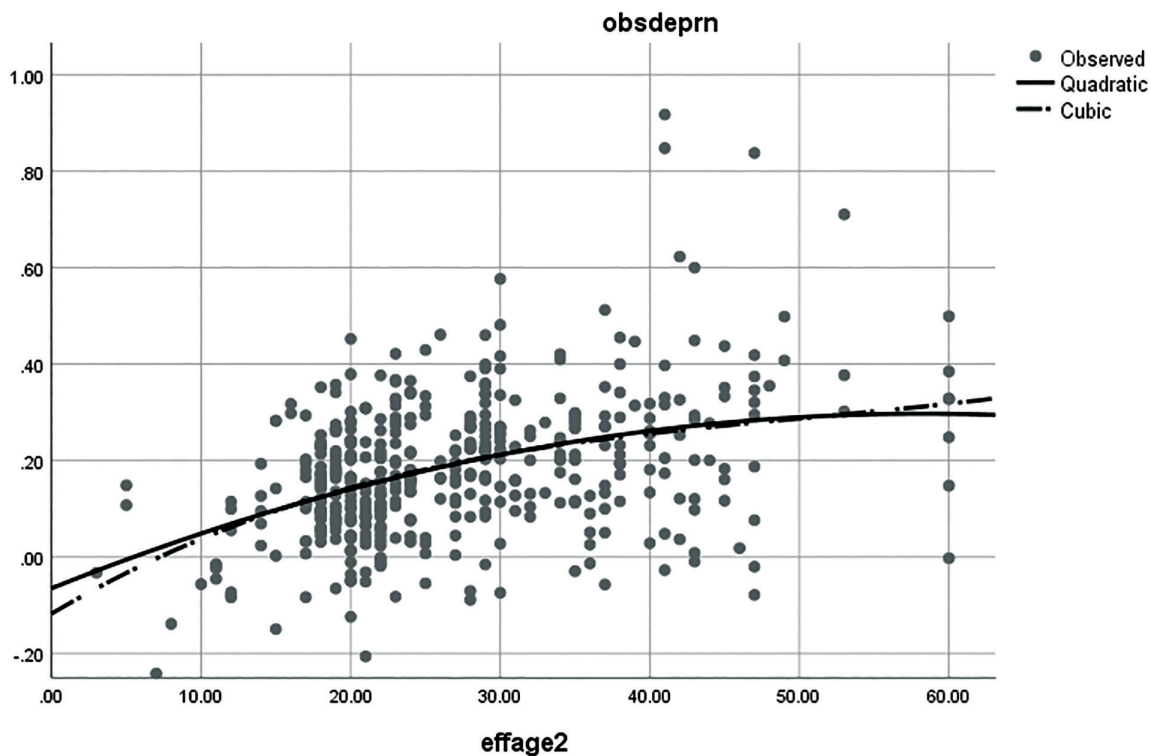
We will now re-run the curve estimation procedure using `effage2` as the independent variable. To use the coefficient results, we again need to increase the number of decimal places shown. This is important since squared and cubed terms can get quite large and using 0.000823 instead of 0.00082344 can introduce inaccuracies and inequities through inadvertent rounding. See Lesson 3 for instructions on expanding the decimal places. The result should appear as follows.

Model Summary and Parameter Estimates

Dependent Variable: `obsdeprn`

Equation	R Square	Model Summary				Parameter Estimates			
		F	df1	df2	Sig.	Constant	b1	b2	b3
Quadratic	.166	44.573	2	447	.000	-.06476010	.01242058	-.00010660	
Cubic	.167	29.896	3	446	.000	-.11737796	.01836116	-.00030660	.00000202

The independent variable is `effage2`.



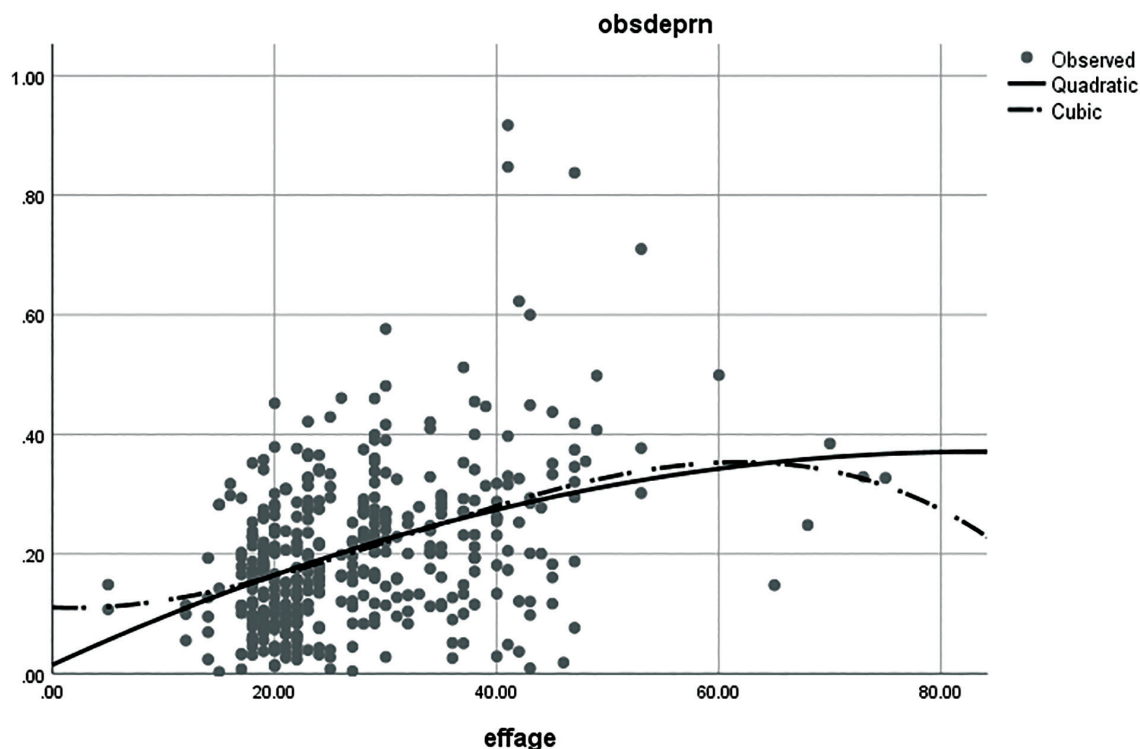
The two curves are almost identical in shape, but both show negative constants indicating an increase in value for new homes rather than the expected value of zero. We will re-run this curve fit with a filter to remove `obsdeprn` values that are less than zero and use the uncapped `effage` as the independent variable.

Model Summary and Parameter Estimates

Dependent Variable: obsdeprn

Equation	R Square	Model Summary				Parameter Estimates			
		F	df1	df2	Sig.	Constant	b1	b2	b3
Quadratic	.163	39.596	2	408	.000	.015	.009	-5.098E-5	
Cubic	.166	26.985	3	407	.000	.111	-.001	.000	-2.244E-6

The independent variable is effage.



The cubic curve has an unacceptable shape. The quadratic is slightly flatter than desirable, but can be accepted without further adjustment. The coefficient table should be expanded to include more decimal places as was done previously.

Model Summary and Parameter Estimates

Dependent Variable: obsdeprn

Equation	R Square	Model Summary				Parameter Estimates			
		F	df1	df2	Sig.	Constant	b1	b2	b3
Quadratic	.163	39.596	2	408	.000	.01467874	.00852441	-.00005098	
Cubic	.166	26.985	3	407	.000	.11069562	-.00076418	.00021458	-.00000224

The independent variable is effage.

We will use the quadratic equation for our depreciation curve. While the cubic has a better R^2 value, the quadratic has the lowest valued constant that represents the depreciation for new homes. As such the depreciation will be created using the following formula:

$$\text{depreciation} = 0.01467874 + (0.00852441 \times \text{effage}) - (0.00005098 \times \text{effage}^2)$$

To apply the depreciation more easily in our cost model, we will create a variable called PCTGOOD (percentage good) that will, in effect, be the opposite of the depreciation. If a home has depreciated 5%, the percent good will be 95%; PCTGOOD is thus created by subtracting the depreciation curve from 1.00. PCTGOOD can be used as a multiplier for the RCN value to determine the current depreciated replacement cost. As mentioned previously, we increased the number of decimal places in the table because the depreciation equation is very sensitive

to the coefficients in the higher power terms. We will use all of the decimal places in this transformation.

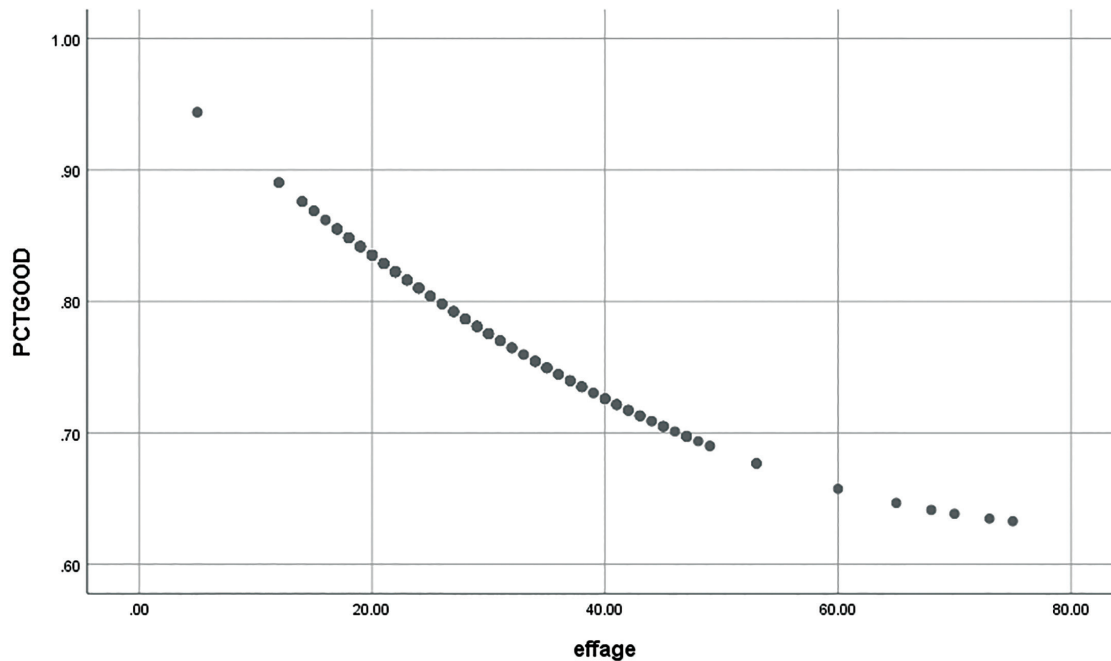
To ensure our upcoming analysis is based on all records, we will first remove the filter (Data → Select Cases → All Cases → OK).

Run the following transformation:

COMPUTE PCTGOOD = 1 - (0.01467874 + 0.00852441 * effage - 0.00005098 * effage**2).

It is good practice to test the shape of this curve to ensure it conforms to expectations. We will produce a scatterplot using Graphs → Legacy dialogs → Scatterplot → Simple Scatter, Define, with PCTGOOD on the vertical axis (Y) and effage on the horizontal axis (X).

The general shape of the curve is what is wanted for our depreciation curve.



You can review the data to see PCTGOOD values varying from 63% to 96% – in other words, a minimum depreciation of 4% and a maximum of 37%.

Other Transformations

At this point, to complete the RCN calculation we need the actual cost data from a cost manual. Once the cost data is available, we can determine the needed transformations to correctly use the information. Here is a summary of the cost data that we will be using in this lesson:

Item	Categories, Costs, and Notes	
Location Adjustment Factor	Not Provided	to be determined using market evidence
Basement	Constant Area rate Finished area	included in 1 storey constant \$21.25 add \$15.60 per square foot
Crawl space	Constant Area rate	included in 1 storey constant \$14.90
1 Storey	Constant 1 storey area rate	if full height basement \$51,500 if crawl \$45,500 \$42.50
2 Storey	Constant 1 storey area rate 2 storey area rate	Add \$16,300 to 1 storey rate \$42.50 \$33.40

continued next page

Item Categories, Costs, and Notes		
Heating	Types 1-6 Hot water heat	included in constant First storey: \$4.40 per square foot Second storey: add \$3.30 per square foot
Fireplaces		\$3,850.00
Baths (full)		\$4,000.00
Baths (partial)		fraction times the full bath rate
Main Building Manual Class	Factor	
40, 41, 42	0.507	
50, 51, 52	0.767	
80, 81, 82	0.783	
90, 91, 92	0.945	
140, 141, 142	1.000	
145, 146, 147	1.220	
150, 151, 15	1.190	
Garage and Carport Manual Class	Factor	
910, 911	0.450	
920, 921	1.000	
930, 931	1.200	
940, 941	1.420	
Carport	Constant Area rate	\$5,850.00 \$22.00
Attached Garage	Constant Area rate	\$11,600.00 \$30.00
Detached Garage	Constant Area rate	\$16,300.00 \$30.00
Basement Garage	Lump Sum	\$7,100.00

There are a number of transformations needed to use this information. We have already included the main building quality factors in `linmcls` and `linmcdv`. In order to deal with the two different constant values for the first floor (depending on full height basement or crawlspace), a transformation for crawlspace area will be needed. The database has been checked and there are no sales with both crawlspace area and full height basement. A crawlspace binary variable will be created so that the two constants can be used. The following transformations will correctly assign the various building constants for the basement type and also add for the additional constant needed for second floor finish.

```

COMPUTE crawlfl = cowlarea.
RECODE crawlfl (0 = 0)(else = 1).
COMPUTE sty1constant = crawlfl.
RECODE sty1constant (0 = 51500)(else = 45500).
COMPUTE sty2fl = flrarea2.
RECODE sty2fl ( 0 = 0)(else = 1).
COMPUTE genconstant = sty1constant + (sty2fl*16300).

```

The next transformations are needed to handle the added cost of hot water heat.

```

COMPUTE supheat = heattype.
RECODE supheat (7,8 = 1)(else = 0).

```

```
COMPUTE heatwater = (supheat *(4.4*flrarea1))+(supheat *(3.3*flrarea2)).
```

The garage and carport quality factors must be created.

```
COMPUTE gr_qfac = gr_mancl.
```

```
RECODE gr_qfac (910 = .45 )(920 = 1)(930 = 1.20)(940 = 1.42)(else=1).
```

```
COMPUTE cp_qfac = cp_mancl.
```

```
RECODE cp_qfac (911 = .45 )(921 = 1)(931 = 1.20)(941 = 1.42)(else=1).
```

Each garage type has a different rate and constant, so separate binary variables are needed for each type.

```
COMPUTE Attgar = gar_type.
```

```
RECODE attgar (1 = 1)(else =0).
```

```
COMPUTE detgar = gar_type.
```

```
RECODE detgar (2 = 1)(else =0).
```

```
COMPUTE bsmtgar = gar_type.
```

```
RECODE bsmtgar (3 = 1)(else =0).
```

All the variables are now in the appropriate format for this model. We can now proceed to develop improvement value.

Developing Improvement Value

The transformation for the improvement RCN less depreciation (IMPRCNLD) can now be created. Normally the first factor in this formula would be the local cost multiplier, to account for local variations from the values in the cost manual. However, we do not yet know the local cost multiplier, and will instead determine it in a subsequent step. The imprcnld equation is as follows:

```
COMPUTE imprcnld = PCTGOOD*(linmcdv*(genconstant+(42.50*flrarea1)+(33.40*flrarea2)
+21.25*bsmtarea+14.90*crwlaarea+15.60*bsmtfin+heatwater+4000*baths+3850*fireplcs)
+cp_qfac*carptfl*(5850+22.00*cp_area)+gr_qfac*garagefl*(attgar*(11600+30.00*gar_area)
+detgar*(16300+30.00*gar_area)+(bsmtgar*7100)))+ob1mktvl.
```

Area rates and constants have been used in an appropriate manner for each cost component. The outbuildings did not have any cost data but an estimate of current market value was provided in the database. Some of these additional values were reasonably large. It is possible that this value should instead have been added later, after all market adjustments were made to the RCNLD. However, most sales do not have any outbuildings so adding the value in at this point will not make a significant difference.

We will now determine the appropriate local cost multiplier, using improvement residuals to represent the market value for the improvements. We create a ratio of imprrsid/imprcnld:

Ratio Statistics for imprrsid / imprcnld

Mean		.721
95% Confidence Interval for Mean	Lower Bound	.708
	Upper Bound	.734
Median		.715
95% Confidence Interval for Median	Lower Bound	.704
	Upper Bound	.728
	Actual Coverage	95.7%
Minimum		.087
Maximum		1.380
Coefficient of Dispersion		.136

The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level. Other confidence intervals are constructed by assuming a Normal distribution for the ratios.

This shows many outliers (minimum is only 0.087), so we will rely on the median value as the better indicator of the adjustment needed. This shows that the `imprcnld` must be reduced, on average, by 0.715 to achieve a median comparable to that of the `imprrsid` – in other words, a 28.5% decrease from the cost manual. We will create a new variable `adjrcnld` to account for this local cost multiplier, by multiplying `imprcnld` by 0.715.

COMPUTE `adjrcnld` = `imprcnld` * 0.715.

We then re-test the new adjusted RCNLD value with “after adjustment” ratio statistics.

Ratio Statistics for <code>imprrsid</code> / <code>adjrcnld</code>		
Mean		1.008
95% Confidence Interval for Mean	Lower Bound	.990
	Upper Bound	1.026
Median		1.000
95% Confidence Interval for Median	Lower Bound	.985
	Upper Bound	1.018
	Actual Coverage	95.7%
Minimum		.121
Maximum		1.929
Coefficient of Dispersion		.136
The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level. Other confidence intervals are constructed by assuming a Normal distribution for the ratios.		

The median is now 1.000, indicating the cost manual data has been correctly adjusted to the `imprrsid` market indication. It appears the cost manual did not directly relate to local or current costs, but now the correct local cost factor has been applied.

Testing Building Characteristics

Further testing will be done to review how well this model values various building characteristics. As we are interested in how well our RCNLD value is predicting the improvement residual, we will base our tests on a new variable called `mktfact1`, equal to `imprrsid` ÷ `adjrcnld`, which you should create now.

COMPUTE `mktfact1` = `imprrsid`/`adjrcnld`.

The first tests will be based on the components of the main building quality class. This variable contains information not only on quality but also on storey height. For example, the first two digits of a 141 class indicate a building class of 14 while the final digit specifies a 1½ storey house. There is a relationship between class and storey because different storey heights can be found among all the quality groups. So, we will test and adjust for storey height first.

The following transformations are needed to extract storey height from the variable `mbmancls`. The MOD transformation returns the remainder when the variable is divided by the number specified, in our case ten, so, for example, the results would be 0 for 80, 1 for 91, 2 for 142, and 5 for 145. The last transformation is needed to capture the storey values for classes 145 to 147.

COMPUTE `storey` = MOD(`mbmancls`,10).

COMPUTE `storey1` = `storey`.

RECODE `storey1` (5 = 0)(6 = 1)(7 = 2).

A Kruskal-Wallis test will be used to determine if all the groups are valued at the same level.

Ranks			
	storey1	N	Mean Rank
mktfact1	.00	415	221.62
	1.00	18	270.22
	2.00	17	272.88
	Total	450	

Test Statistics^{a,b}

mktfact1	
Kruskal-Wallis H	4.755
df	2
Asymp. Sig.	.093

a. Kruskal Wallis Test

b. Grouping Variable:
storey1

The Asymptotic Significance (Asymp. Sig.) value is greater than 0.05, so the test does pass at the 95% confidence level. However, the mean ranks for storey1 values 1 and 2 are much greater than the value for group 0, so we will run ratio statistics.

Ratio Statistics for imprrsid / adjrcnld

Group	Median	95% Confidence Interval for Median		Actual Coverage	Minimum	Maximum	Coefficient of Dispersion
		Lower Bound	Upper Bound				
.00	.993	.982	1.015	95.1%	.121	1.929	.137
1.00	1.063	.906	1.237	96.9%	.701	1.500	.168
2.00	1.056	1.005	1.090	95.1%	.614	1.222	.085
Overall	1.000	.985	1.018	95.7%	.121	1.929	.136

The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level.

This shows that the median values for both Group 1 and Group 2 are greater than 5% above the desired level of 1.000, so adjustments are needed for these groups. The adjustments will be made with the following transformations:

COMPUTE storeyfact = storey1.

RECODE storeyfact (1 = 1.063)(2=1.056)(else =1).

COMPUTE storeyrcnld = storeyfact*adjrcnld.

The “after adjustment” ratio statistics show that the adjustments have been applied correctly as the medians for the adjusted Groups are now equal to 1.00.

Ratio Statistics for imprrsid / storeyrcnld

Group	Median	95% Confidence Interval for Median		Actual Coverage	Minimum	Maximum	Coefficient of Dispersion
		Lower Bound	Upper Bound				
.00	.993	.982	1.015	95.1%	.121	1.929	.137
1.00	1.000	.853	1.163	96.9%	.660	1.412	.168
2.00	1.000	.951	1.032	95.1%	.581	1.157	.085
Overall	.994	.980	1.010	95.7%	.121	1.929	.136

The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level.

We now need a new factor is needed to test quality.

COMPUTE mktfact2 = imprrsid / storeyrncld.

As the next table shows, in order to properly test the quality of the building, it will be necessary to group some of the classes as many do not have enough sales for individual analysis.

Main building class (quality and story)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	40.00	11	2.4	2.4	2.4
	41.00	2	.4	.4	2.9
	50.00	1	.2	.2	3.1
	51.00	1	.2	.2	3.3
	80.00	41	9.1	9.1	12.4
	81.00	15	3.3	3.3	15.8
	90.00	189	42.0	42.0	57.8
	92.00	3	.7	.7	58.4
	140.00	169	37.6	37.6	96.0
	142.00	9	2.0	2.0	98.0
	145.00	1	.2	.2	98.2
	150.00	3	.7	.7	98.9
	152.00	5	1.1	1.1	100.0
	Total	450	100.0	100.0	

The grouping is done with the following transformations:

COMPUTE qualcls = mbmancls.

RECODE qualcls (30 THRU 42=4)(50 THRU 82=8)(90 THRU 92=9)(140 THRU 142=14)
(145 THRU HIGHEST=15).

Classifications from 145 to 152 are grouped together as qualcls 15; classifications 140 through 142 are grouped as qualcls 14; 90 through 92 as 9, 50 through 82 as 8, and the lowest classifications, those less than 50, are grouped as qualcls 4. This will give us sufficient counts within the qualcls variable if we need to make adjustments to the storey adjusted rcnld variable storeyrncld.

A K-W test was done using the new factor and qualcls with the range defined as 4 to 15.

Ranks

	qualcls	N	Mean Rank
mktfact2	4.00	13	248.85
	8.00	58	261.26
	9.00	192	213.39
	14.00	178	226.48
	15.00	9	200.33
	Total	450	

Test Statistics^{a,b}

mktfact2	
Kruskal-Wallis H	6.817
df	4
Asymp. Sig.	.146

a. Kruskal Wallis Test

b. Grouping Variable:
qualcls

The Asymp. Sig. is greater than 0.05, indicating we cannot reject the null hypothesis that all groups are equally valued. However, the mean ranks for qualcls = 4 and qualcls = 8 are a little high and the mean rank for qualcls = 15 it is a bit low. A ratio statistics report will be run to determine if any adjustments are warranted.

Ratio Statistics for imprrsid / storeyrcnld

Group	Median	95% Confidence Interval for Median		Actual Coverage	Minimum	Maximum	Coefficient of Dispersion
		Lower Bound	Upper Bound				
4.00	1.061	.584	1.703	97.8%	.323	1.929	.336
8.00	1.057	.998	1.130	95.2%	.121	1.833	.220
9.00	.980	.946	1.008	96.4%	.601	1.315	.120
14.00	.995	.978	1.018	95.7%	.581	1.479	.106
15.00	.961	.932	1.032	96.1%	.769	1.143	.064
Overall	.994	.980	1.010	95.7%	.121	1.929	.136

The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level. Other confidence intervals are constructed by assuming a Normal distribution for the ratios.

As we noted with the values for the mean ranks; for qualcls = 4 and qualcls = 8, the median level is high and will need to be adjusted. For qualcls = 15, the median level is a little low, but it is within 5% of 1.000 and its confidence interval brackets 1.000; so, no adjustment is needed. The adjustments for qualcls = 4 and qualcls = 8 will be made with the following transformations:

COMPUTE qualfact=qualcls.

RECODE qualfact (4=1.061)(8=1.057)(else=1).

COMPUTE qualrcnld=qualfact*storeyrcnld.

COMPUTE mktfact3=imprrsid/qualrcnld.

The “after adjustment” ratio statistics show that the adjustments have been applied correctly as the medians for the adjusted Groups are now equal to 1.00.

Ratio Statistics for imprrsid / qualrcnld

Group	Median	95% Confidence Interval for Median		Actual Coverage	Minimum	Maximum	Coefficient of Dispersion
		Lower Bound	Upper Bound				
4.00	1.000	.550	1.606	97.8%	.304	1.819	.336
8.00	1.000	.944	1.069	95.2%	.115	1.734	.220
9.00	.980	.946	1.008	96.4%	.601	1.315	.120
14.00	.995	.978	1.018	95.7%	.581	1.479	.106
15.00	.961	.932	1.032	96.1%	.769	1.143	.064
Overall	.988	.974	1.004	95.7%	.115	1.819	.133

The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level.

Next splitlvl will be tested with a Mann-Whitney test.

Ranks

	Binary - home is split level	N	Sum of Ranks	
			Mean Rank	
mktfact3	.00	418	222.06	92821.00
	1.00	32	270.44	8654.00
	Total	450		

Test Statistics^a

mktfact3	
Mann-Whitney U	5250.000
Wilcoxon W	92821.000
Z	-2.028
Asymp. Sig. (2-tailed)	.043

a. Grouping Variable: Binary -
home is split level

The test does not pass at the 95% confidence level because the Asymp. Sig. value of 0.043 is less than 0.05. So, the ratio statistics will be checked.

Ratio Statistics for imprrsid / qualrcnld

Group	Median	95% Confidence Interval for Median		Actual Coverage	Minimum	Maximum	Coefficient of Dispersion
		Lower Bound	Upper Bound				
.00	.985	.970	1.000	95.5%	.115	1.819	.133
1.00	1.052	.955	1.139	98.0%	.581	1.395	.128
Overall	.988	.974	1.004	95.7%	.115	1.819	.133

The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level.

This shows a median for the split-level group that is more than 5% greater than 1.00. As such, an adjustment is needed. As well, for the non split-level properties, 1.000 is not in between the lower and upper bounds of the 95% confidence interval around the median; so, an adjustment is required here as well. The following transformations are needed to perform and apply the adjustment.

```
COMPUTE spltfact = splitlvl.
RECODE spltfact (0=.985)(1=1.052).
COMPUTE spltrcnld = spltfact * qualrcnld.
COMPUTE mktfact4 = imprrsid/spltrcnld.
```

The “after adjustment” ratio statistics show that the adjustment was applied successfully.

Ratio Statistics for imprrsid / spltrcnld

Group	Median	95% Confidence Interval for Median		Actual Coverage	Minimum	Maximum	Coefficient of Dispersion
		Lower Bound	Upper Bound				
.00	1.000	.985	1.016	95.5%	.117	1.846	.133
1.00	1.000	.908	1.083	98.0%	.553	1.326	.128
Overall	1.000	.984	1.016	95.7%	.117	1.846	.132

The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level.

Another variable that we should test in this stage is the garage type. This is necessary because the cost for detached garage has a larger constant than that for an attached garage, and in a cold climate it is likely that an attached garage would be the preferred garage type. We will carry

out a K-W test, with mktfact4 as the Test Variable and gar_type as the Grouping Variable (with a Defined Range of 0 to 3).

Ranks			
	Garage type	N	Mean Rank
mktfact4	.00	338	221.96
	1.00	39	236.85
	2.00	46	189.35
	3.00	27	314.96
	Total	450	

Test Statistics^{a,b}

mktfact4	
Kruskal-Wallis H	16.879
df	3
Asymp. Sig.	.001

a. Kruskal Wallis Test

b. Grouping Variable:
Garage type

Garage type fails at the 95% confidence level because the Sig. value is less than 0.05. This shows an adjustment is likely to be required. A quick scan of the mean ranks confirms this as type 2 (detached) and type 3 (basement) are not very close to the target value of 225. Ratio statistics will show what adjustments, if any, are needed.

Ratio Statistics for imprrsid / splitrcnld

Group	Median	95% Confidence Interval for Median		Actual Coverage	Minimum	Maximum	Coefficient of Dispersion
		Lower Bound	Upper Bound				
.00	.997	.978	1.015	95.6%	.117	1.815	.134
1.00	1.010	.960	1.054	97.6%	.738	1.281	.096
2.00	.973	.886	1.025	97.4%	.559	1.846	.157
3.00	1.083	1.024	1.212	98.1%	.952	1.408	.099
Overall	1.000	.984	1.016	95.7%	.117	1.846	.132

The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level.

For basement garages the median is 8.3% greater than 1.000, and the 95% confidence interval does not bracket 1.000. However, for attached garages, the median is between 0.95 and 1.05 and the 95% confidence interval brackets 1.000. As such, only basement garages require adjusting. The following transformations are needed to perform and apply the adjustment.

COMPUTE garfact = gar_type.

RECODE garfact (3=1.083)(else=1).

COMPUTE garcnld = garfact*splitrcnld.

COMPUTE mktfact5 = imprrsid/garcnld.

A new set of ratio statistics will confirm that the adjustment was applied correctly.

Ratio Statistics for imprrsid / garcnld

Group	Median	95% Confidence Interval for Median		Actual Coverage	Minimum	Maximum	Coefficient of Dispersion
		Lower Bound	Upper Bound				
.00	.997	.978	1.015	95.6%	.117	1.815	.134
1.00	1.010	.960	1.054	97.6%	.738	1.281	.096
2.00	.973	.886	1.025	97.4%	.559	1.846	.157
3.00	1.000	.945	1.119	98.1%	.879	1.300	.099
Overall	.997	.980	1.008	95.7%	.117	1.846	.131

The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level.

We will run a few more tests on the new ratio variable mktfact5. Kruskal-Wallis tests were run for the discrete variables and Mann-Whitney tests for the binary variables. The results are summarized in the table below.

Variable	Asymp. Sig.
carptfl	0.525
crawlfl	0.152
fireplcs	0.925
bedrooms	0.732
familyrm	0.442
baseste	0.113
bath100	0.465
gr_mancl	0.891
cp_mancl	0.628
sup_heat	0.177

Please note that the Kruskal-Wallis test will not run correctly when using a Grouping variable that has any decimal places; the values of the variable will be rounded to the nearest integer. For that reason a transformation had to be run on the bathroom count variable to remove the decimal places. The syntax statements below eliminate the decimal places from the bathroom variable making the 1 a 100, the 1.25 a 125, the 1.5 a 150 and so forth. The name of the new variable is bath100.

```
COMPUTE bath100 = baths.
```

```
RECODE bath100 (1=100) (1.5=150) (1.75=175) (2=200) (2.25=225) (2.5=250) (ELSE=275).
```

```
EXECUTE.
```

All of the Asymptotic Significance values are greater than 0.05. As such, it appears that no further adjustments are needed to our RCNLD value.

That completes the testing of the RCNLD value against the improvement components. The final adjusted improvement value, garcnld, can be added to the adjusted land value to calculate an overall market value under the cost approach.

At this point a short recap might be in order. We have a final cost value to calibrate that will be done in the next few pages. How did we get here? First, we created an improvement residual value (imprrsid) that was based on the time adjusted sale price and the land value created in the previous lesson. We then determined the best depreciation curve for the improvements in our database (obsdeprn) and calculated a factor we could apply directly to represent this depreciation (pctgood).

We then created the new variables we needed to properly apply the cost data. Once this was done, we built the costing formula to create the new variable imprcnld. We then tested this variable to find a revised local cost adjustment factor (0.715). Following that we tested the newly adjusted depreciated improvement cost value (adjrcnld) against a variety of building related variables to ensure our RCNLD value was equitable across those variables, this included storey, qualcls, splitlvl, and

gar_type. We used ratio statistics at each stage to determine whether an adjustment was needed, and, if one was needed, what the adjustment should be. This provided us with a final improvement value that is uniform across a number of improvement characteristics. We will now move on to computing overall market values under the cost approach for the properties in the 444costdata database.

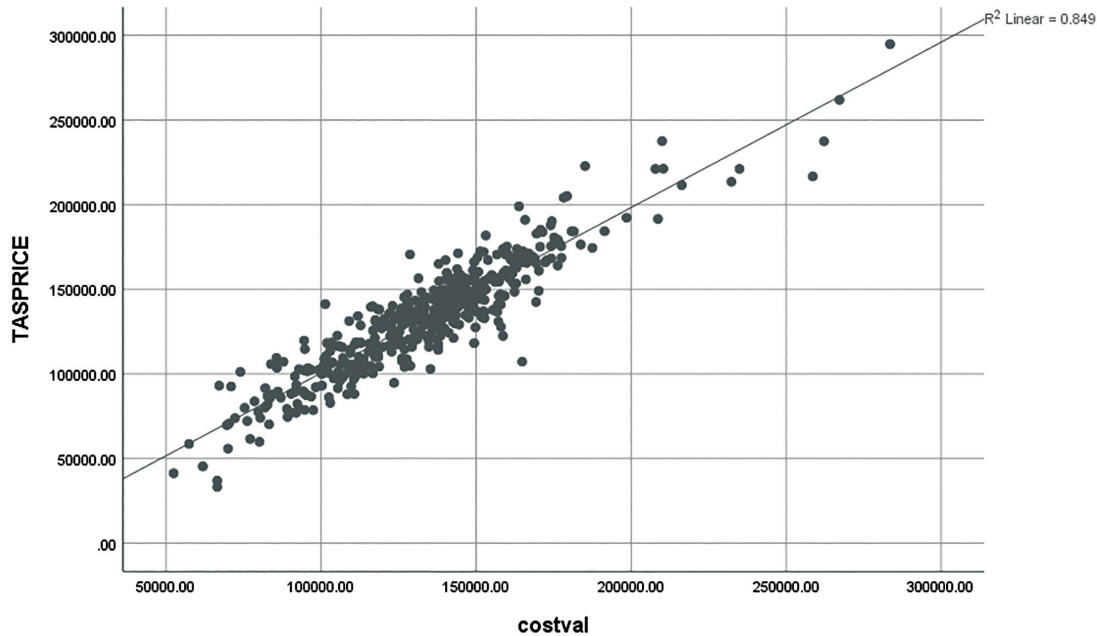
Steps 6, 7, and 8 – Model Calibration and Testing

Now that we have an improvement value (garcnld) and a land value (neighnld from Lesson 3), we are now ready to calibrate the model. In a cost model, calibration simply means creating a transformation to apply costs to the variables of interest. We will also create an ASR for the cost values generated, in order to begin testing these results.

COMPUTE costval = garcnld + neighnld.

COMPUTE costasr = costval / tasprice.

We first examine a scatterplot of costval compared to tasprice:



The R^2 of 0.849 shows a very good relationship. Next, the ratio statistics for costasr will be examined.

Ratio Statistics for costval / TASPRICE

Mean		1.014
95% Confidence Interval for Mean	Lower Bound	1.003
	Upper Bound	1.025
Median		1.002
95% Confidence Interval for Median	Lower Bound	.995
	Upper Bound	1.012
	Actual Coverage	95.7%
Minimum		.718
Maximum		2.000
Coefficient of Dispersion		.081

The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level. Other confidence intervals are constructed by assuming a Normal distribution for the ratios.

This shows an excellent median and a mean only slightly greater. The 8.1% COD meets the IAAO's standard of less than 10%. Next, we will analyze the valuation by neighbourhood.

Neighbourhood Analysis

We first run a Kruskal-Wallis test on the costasr to determine if the neighbourhoods have been equitably valued.

Ranks			
	Sub neighbourhood	N	Mean Rank
costasr	713.00	80	217.24
	715.00	56	251.96
	716.00	159	235.19
	731.00	27	136.44
	732.00	12	196.42
	734.00	21	395.86
	736.00	30	199.97
	737.00	65	188.26
	Total	450	

Test Statistics^{a,b}

costasr	
Kruskal-Wallis H	59.309
df	7
Asymp. Sig.	.000

a. Kruskal Wallis Test

b. Grouping Variable:
Sub neighbourhood

This shows that at the 95% level we can reject the null hypothesis that the neighbourhoods are equally valued; the Sig. value is 0.000. Ratio statistics will be used to determine what adjustments, if any, are needed.

Ratio Statistics for costval / TASPRICE

Group	Median	95% Confidence Interval for Median		Actual Coverage	Minimum	Maximum	Coefficient of Dispersion
		Lower Bound	Upper Bound				
713.00	.999	.968	1.025	96.7%	.718	1.362	.091
715.00	1.025	1.000	1.064	95.6%	.754	1.538	.085
716.00	1.004	.995	1.020	96.1%	.831	1.314	.053
731.00	.901	.833	.994	98.1%	.767	1.196	.117
732.00	.978	.891	1.075	96.1%	.823	1.142	.081
734.00	1.184	1.091	1.240	97.3%	1.002	2.000	.112
736.00	.985	.936	1.052	95.7%	.838	1.148	.067
737.00	.973	.941	1.000	95.4%	.868	1.338	.069
Overall	1.002	.995	1.012	95.7%	.718	2.000	.081

The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level. Other confidence intervals are constructed by assuming a Normal distribution for the ratios.

This shows the median for neigh 734 is far too high and the median for 731 is too low. The remaining medians are within 5% of 1.000, however, the 95% confidence intervals around the median for neighbourhoods 715 and 737 do not bracket 1.000 (in one case the lower bound is

equal to 1.000 and the in the other case the upper bound is equal to 1.000). This means that adjustments are required for neighbourhoods 715, 731, 734 and 737. You should note that, in this case, we are adjusting the numerator of the ratio (costval), as such, we will be using the reciprocal of the median to make the adjustments (i.e., dividing 1 by the median value):

COMPUTE neighfact = neigh.

RECODE neighfact (715=0.9756)(731=1.1099)(734=0.8446)(737=1.0277)(else=1).

COMPUTE ncostval = neighfact * costval.

COMPUTE ncostasr = ncostval / tasprice.

Re-running ratio statistics provides the following:

Ratio Statistics for ncostval / TASPRICE

Group	Median	95% Confidence Interval for Median		Actual Coverage	Minimum	Maximum	Coefficient of Dispersion
		Lower Bound	Upper Bound				
713.00	.999	.968	1.025	96.7%	.718	1.362	.091
715.00	1.000	.976	1.038	95.6%	.736	1.500	.085
716.00	1.004	.995	1.020	96.1%	.831	1.314	.053
731.00	1.000	.924	1.104	98.1%	.851	1.327	.117
732.00	.978	.891	1.075	96.1%	.823	1.142	.081
734.00	1.000	.921	1.048	97.3%	.846	1.689	.112
736.00	.985	.936	1.052	95.7%	.838	1.148	.067
737.00	1.000	.967	1.028	95.4%	.892	1.375	.069
Overall	1.001	.994	1.012	95.7%	.718	1.689	.075

The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level. Other confidence intervals are constructed by assuming a Normal distribution for the ratios.

This shows all neighbourhood medians are at an acceptable level and only neighbourhoods 731 and 734 having a COD greater than the 10% threshold for homogeneous neighbourhood standard set by the IAAO. However, they are within IAAO's 15% level for heterogeneous neighbourhoods.

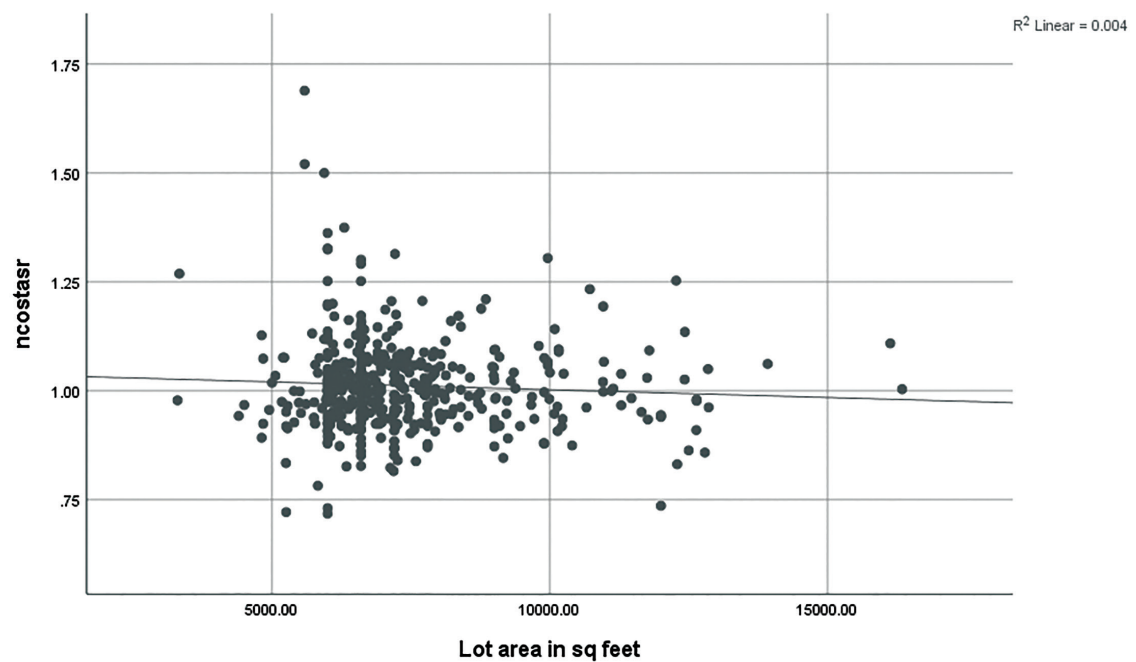
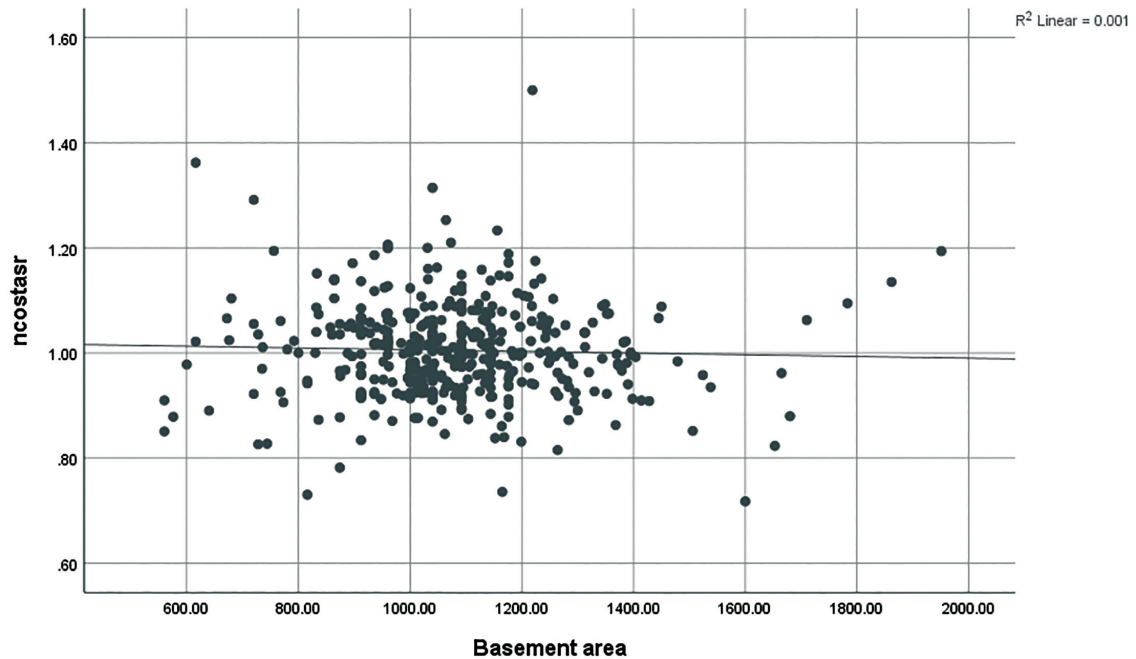
Ratio Statistics for ncostval / TASPRICE

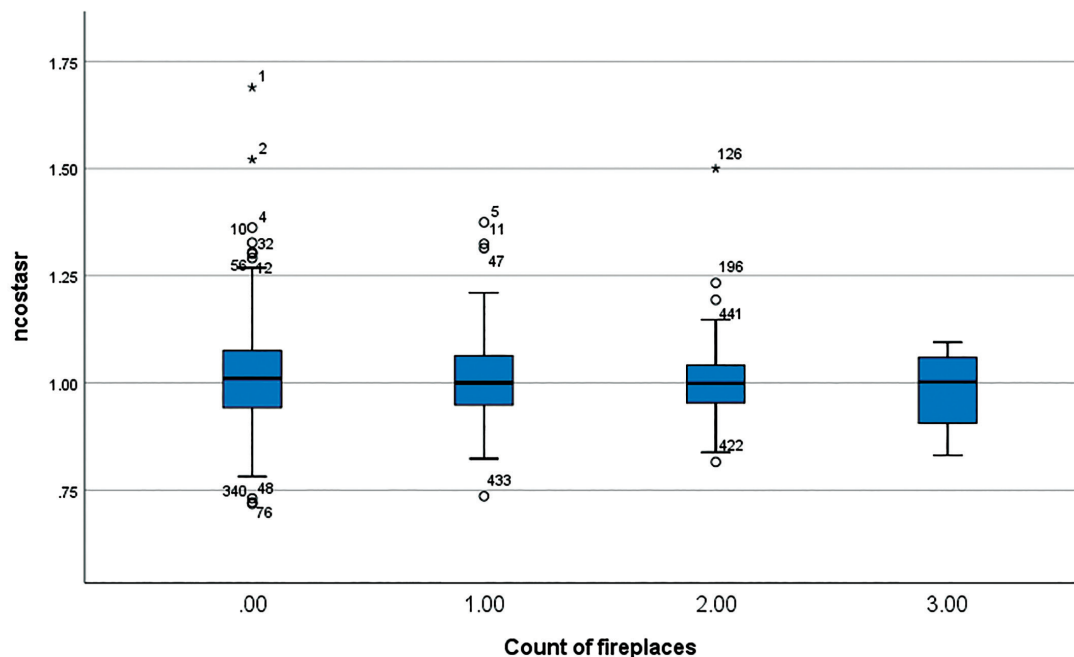
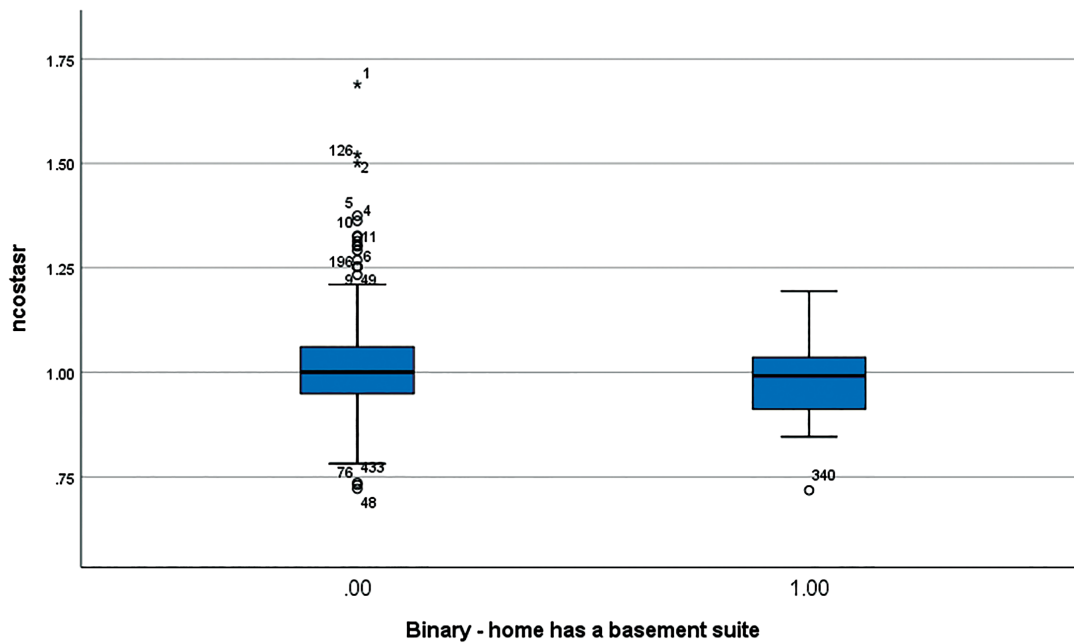
Mean		1.012
95% Confidence Interval for Mean	Lower Bound	1.002
	Upper Bound	1.022
Median		1.001
95% Confidence Interval for Median	Lower Bound	.994
	Upper Bound	1.012
	Actual Coverage	95.7%
Minimum		.718
Maximum		1.689
Coefficient of Dispersion		.075

The confidence interval for the median is constructed without any distribution assumptions. The actual coverage level may be greater than the specified level. Other confidence intervals are constructed by assuming a Normal distribution for the ratios.

This shows a good overall result with the median equal to 1.001 and the mean slightly greater than that at 1.012. The COD is 7.5%, within the IAAO standard for homogeneous neighbourhoods.

For our purposes, this completes the cost model; but, in practice, we would continue testing the costasr to look for places where there is inequality in the model and, if any are discovered, make the appropriate adjustments. However, since we have outlined the testing process in both the land and improvement portions of the cost model development, we will not subject the reader to another round of K-W and M-W tests, ratio statistics tests, and adjustment transformations. Instead, we will show only a few graphics: samples of scatterplots and boxplots showing no relationship between the final ASR and two continuous and two discrete variables:





SUMMARY

In this lesson, you have:

- stated a general cost model;
- examined the available variables in the database;
- developed a market-derived depreciation curve;
- created an improvement value using cost data provided;
- tested and adjusted the improvement value to correspond to the local market;
- created a cost value using the market adjusted improvement value, together with the land value developed in Lesson 3; and
- performed a few tests on the final result.

This completes the cost approach portion of the course. Project 1 will give you the opportunity to apply the techniques presented in Lessons 1 through 4 in a practical example. The next lessons will focus on the use of regression techniques for sales comparison valuation methods.

REVIEW AND DISCUSSION QUESTIONS



The following questions are for review and discussion purposes only. They are not to be submitted as part of the assignment for this lesson. Answers to select questions can be found on the Course Resources webpage. However, in order to develop your understanding, you are encouraged to attempt answering the questions before accessing the answers. You may also wish to discuss these questions with your fellow students on the course discussion forum – through discussing these questions, you will better understand the course materials and be better prepared for the final examination.

1. What are the main advantages and disadvantages to using cost models in residential assessment work?
2. Why might assessment staff have an easier time than single property appraisers in developing depreciation estimates?
3. Define the following terms and explain how they relate to the cost modeling demonstrated in this lesson:
 - Physical Life
 - Economic Life
 - Effective Age
 - Actual Age
 - Remaining Economic Life
4. What is the advantage to using replacement cost compared to reproduction cost in a cost model?

ASSIGNMENT 4

Lesson 4: Cost-Based Modeling

Marks: 1 mark per question

1. Step 3 of the general model building process, “Examine the potential independent variables for relationships with each other and with the dependent variable”, can be ignored when building a cost model. Why?
 - (1) There are no independent variables to consider.
 - (2) All possible correlations between variables will be equal.
 - (3) Only the cost values from the manual are of interest.
 - (4) The dependent variable is unknown.
2. Curve estimation is used with a calculated value for observed depreciation in order to establish:
 - (1) an expression for the amount of depreciation for each year.
 - (2) the correlation between depreciation and sale date.
 - (3) the distribution of effective ages in the database.
 - (4) whether or not the age/life method of depreciation was used by the appraiser.
3. Using the curve estimation process in SPSS, depreciation is generally best estimated by which of the following types of relationships?
 - (1) Linear and Exponential
 - (2) Quadratic and Cubic
 - (3) Exponential and Quadratic
 - (4) Cubic and Logarithmic
4. You are using Kruskal-Wallis to test if the ASR given by the cost model is equitably distributed across all six neighbourhoods in a valuation model. If the database has 540 records, what mean rank would you be looking for?
 - (1) 540
 - (2) 90
 - (3) 270
 - (4) 301
5. What is a good next step if you discover that a variable “fails” the Kruskal-Wallis test (that is, you reject the null hypothesis)?
 - (1) Run a compare means test to verify the “failure”.
 - (2) Adjust the variable based on the highest and lowest mean ranks given in the Kruskal-Wallis test.
 - (3) Run a ratio statistics test to verify the “failure”.
 - (4) Remove any high or low outliers and re-run the Kruskal-Wallis test.
6. The chi-squared value calculated for a Kruskal-Wallis test is 17.04. You can conclude:
 - (1) the variable “fails” the Kruskal-Wallis test.
 - (2) the variable “passes” the Kruskal-Wallis test.
 - (3) the variable needs an adjustment of $(1 \div 17.04)$ in order to “pass” the Kruskal-Wallis test.
 - (4) nothing, because the chi-squared value alone is of no use.

Assignment continues on the following page

ASSIGNMENT 4, continued

7. You are developing a cost-based valuation model. You have run a series of ratio statistics on the resulting ASR using a number of grouping variables including neighbourhood, quality of construction, garage type, basement type, and number of stories. You find that adjustments are needed for neighbourhood, quality of construction, and number of stories. What is the correct plan of action?
- (1) Adjust for neighbourhood, quality of construction, and number of stories then re-run ratio studies for those three variables.
 - (2) Adjust for neighbourhood, quality of construction, and number of stories then re-run all parametric normality tests for these three variables.
 - (3) Adjust for any of the three variables first, then re-test the other variables, adjust as needed, and re-test, until no other adjustments are indicated.
 - (4) Use a Kruskal-Wallis test to confirm the need for adjustments, but only adjust if the ASR “fails” the Kruskal-Wallis test. Ratio statistics can be ignored.

THE NEXT THREE (3) QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:

Bojack Valuation is developing a cost valuation model. They want to use a ratio statistics test to examine the resulting ASR's mean, median, and coefficient of dispersion grouped by a number of different variables. However, they need some help interpreting the test results. Consider the following possible test results in answering the three questions below.

- A. the median is greater than 1.05
 - B. the mean is greater than 1.05
 - C. the median is less than 0.95
 - D. the mean is less than 0.95
 - E. the median is between 0.95 and 1.05
 - F. the mean is between 0.95 and 1.05
 - G. the 95% confidence interval around the median brackets 1.00
 - H. the 95% confidence interval around the mean brackets 1.00
 - I. the 95% confidence interval around the median does not bracket 1.00
 - J. the 95% confidence interval around the mean does not bracket 1.00
 - K. the coefficient of dispersion is greater than 0.10
 - L. the coefficient of dispersion is less than 0.15
8. An adjustment to the calculated cost value should be made:
- (1) in all cases except E and F.
 - (2) for cases A, C, and I only.
 - (3) for cases B, D, and J only.
 - (4) for cases E, F, G, H, and K only.
9. No adjustment to the cost value is warranted:
- (1) in all cases.
 - (2) for case E only.
 - (3) for case G only.
 - (4) for cases E and G only.
10. Generally speaking, in a ratio statistics test for a cost model you are looking for:
- (1) cases E, G, and L only.
 - (2) cases F, H, and K only.
 - (3) cases G, H and K only.
 - (4) cases E, F, G, H, and L.

End of assignment

**Planning Ahead**

Project 1 is due in one week. You should be well into the work required for this project by now.

